AI221 Project Proposal

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November 2026

1 Proposed Project Title:

Machine Learning-Based Classification of Thermal Resistance Degradation in Solder Joints Using Void Spatial Distribution Features from X-ray Images

2 Description of the Problem

In the assembly of semiconductor products, devices are typically soldered onto printed circuit boards (PCBs) through a reflow soldering process, wherein the device is exposed to a high temperature to melt the pre-applied solder paste and is then exposed to a lower temperature to let it solidify. During this process, microscopic gas pockets can become trapped within the solder joint, creating what are known as solder voids. These voids are especially critical to detect in Class 3 assemblies, which are typically used in aerospace, automotive, and medical applications, as they can lead to premature failure and reduce product lifespan.

Solder voids are commonly detected after assembly through non-destructive inspection techniques, primarily X-ray imaging. Scanning Acoustic Microscopy (SAM) can also be used, mainly for package-level interfaces or where the board structure allows. Current industry standards and specifications, such as IPC-A-610, IPC-7095, IPC-7093, and MIL-STD-883, evaluate solder joint and die attach acceptability primarily based on the total void area, commonly around 25% for solder joints, although the exact limit varies by component type and customer specification. Some of these documents also address void type or the size of an individual void, but, to the authors' knowledge, none defines location-dependent acceptance limits that relate void position to thermal performance, particularly for thermal pads. This limitation creates a significant reliability concern. A void occupying 20% of the solder area directly beneath a thermal pad can obstruct the primary heat dissipation pathway and potentially cause greater thermal degradation than a similarly sized void distributed near the periphery of the joint. Despite their substantially different thermal impacts, both conditions are assessed largely on the basis of total void percentage. Consequently, the relationship between void spatial distribution and thermal performance remains inadequately addressed by existing acceptance criteria.

This study aims to bridge this gap by investigating whether spatial void distribution characteristics extracted from X-ray and, where applicable, SAM images can be used to predict thermal resistance degradation during reliability testing. This study could bring value by potentially giving early insight into whether a product may fail prematurely, because a released product in manufacturing typically does not undergo reliability testing. This study could potentially be an added screening test, especially for Class 3 assemblies. Lastly, it could also potentially reduce the qualification time for new products and help identify process issues earlier.

3 KeyRelated Literature

1. *Predicting Thermal Resistance of Solder Joints Based on Scanning Acoustic Microscopy Using Artificial Neural Networks.*
`https://ieeexplore.ieee.org/document/9939465`

2. *Comparison of Different Input Data for the Prediction of LED Solder Joints Using Artificial Neural Networks.*
`https://ieeexplore.ieee.org/document/11250977`

3. *Investigations on High-Power LEDs and Solder Interconnects in Automotive Application: Part II – Reliability.*
`https://publica.fraunhofer.de/entities/publication/ecb4127b-b53e-4cf7-a193-9bfc1eeca4eb`

4. Wang, Y. et al., *Effect of Voids on the Thermal Fatigue Reliability of PBGA Solder Joints through Submodel Technology.*
`https://www.researchgate.net/publication/224376390_Effect_of_Voids_on_the_Thermal_Fatigue_Reliability_of_PBGA_Solder_Joints_through_Submodel_Technology`

5. Lau, J. H. et al., *Effect of Voids on the Reliability of BGA/CSP Solder Joints.*
`https://www.sciencedirect.com/science/article/abs/pii/S0026271403001240`

6. Chen, X. et al., *Effects of Voids on Thermal Fatigue Reliability of Solder Joints on Inner Rings in Ball Grid Array Packaging by Finite Element Analysis.*
`https://www.researchgate.net/publication/368951403_Effects_of_Voids_on_Thermal_Fatigue_Reliability_of_Solder_Joints_on_Inner_Rings_in_Ball_Grid_Array_Packaging_by_Finite_Element_Analysis`

7. Zhang, Y. et al., *Segmentation of Void Defects in X-ray Images of Chip Solder Joints Based on PCB-DeepLabV3 Algorithm.*
`https://www.nature.com/articles/s41598-024-61643-w`

For the proposed topic, the first three references are particularly important.

- Fraunhofer reliability study provides the experimental LED reliability and thermal degradation data that originated the Hella/Kaggle dataset.

- Predicting Thermal Resistance of Solder Joints Based on SAM Using ANN demonstrates that machine learning can predict thermal resistance from inspection data.
- Comparison of Different Input Data for the Prediction of LED Solder Joints Using ANN directly compares different inspection modalities (including X-ray and SAM) for reliability prediction.

Reliability Testing → Thermal Resistance Degradation → Inspection Data (X-ray/SAM) → Machine Learning Prediction → Proposed Work: Spatial Void Feature-Based Thermal Performance Classification

3.1 About the Dataset

The LED Solder Joint Reliability dataset (*hellastudy*) is publicly available on Kaggle and contains data from 1,800 high-power LED samples spanning nine LED types and five solder paste formulations. Each sample was inspected and evaluated using three complementary characterization methods.

Table 1: Inspection and Testing Methods Used in the Dataset

Inspection/Test Method	**What It Captures**	**Role in This Project**
X-ray Imaging	Two-dimensional projection of the solder joint showing void locations, sizes, and distributions	Primary input source for extracting spatial void distribution features
Scanning Acoustic Microscopy (SAM)	Cross-sectional visualization of delamination and three-dimensional void structures	Supplementary input source providing depth-related information
Transient Thermal Analysis (TTA)	Measurement of thermal resistance (R_{th}) over time through electrical characterization	Target variable for supervised learning and thermal performance classification

Following inspection, all 1,800 units were subjected to thermal shock cycling for up to 1,500 cycles. Thermal resistance (R_{th}) measurements were recorded periodically throughout the reliability test.
The LED Solder Joint Reliability dataset (*hellastudy*) is publicly available on Kaggle and can be accessed at:

`https://www.kaggle.com/datasets/andreaszippelius/hellastudy-of-leds2/data`

4 Methodology

4.1 Void Spatial Features Extraction and Pre-processing

1. PCA
2. UMAP
3. Isomap
4. t-SNE for visualization

4.2 Unsupervised Learning

1. K-Means
2. DBSCAN

4.3 Supervised Learning

1. Random Forest
2. XGBoost
3. SVM
4. ANN